

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

Governance Log

Audit records, human checkpoints, bias check, guardrails and incidents

AI-driven Project Management System for the Harbourline Community Hub Refurbishment

Course: MA6094 Agentic AI for Project Management
Capstone Project Group 5

Tian Yujia	G2510505D	team lead
Qu Chengzhi	G2510533D	
Zhao Lingfeng	G2510595H	
Zhang Yinan	G2505943K	
Liu Beiqi	G2510916G	
Huang Hao	G2510626K	

Document owner: Huang Hao

14 November 2026

Abstract

This log reports the governance records that the Harbourline AI PM system writes about itself into its SQLite tables `audit_log`, `approvals` and `handoffs`. The evidence is a scripted demonstration run on the latest code, in baseline mode against the mock board, with 188 audit rows, 137 agent writes and 10 approvals, plus a check of the start-screen Reset demo and a read-only snapshot of the team's Jira project. All 92 PM issues have a `pm create` row, and no agent wrote to the PM tool before checkpoints HC2 and HC3 were approved. Approvals were signed by seeded role accounts; they are demonstration decisions, not evidence of a real person's sign-off. Refused attempts were blocked and logged: a Group Leader approving HC2, release of a Critic-stopped draft, and agent close, delete and Done attempts. The bias and fairness check BIAS-1 found six skews in the 214 consultation responses, three of them high severity. The team carried out four of its five actions but not the second multilingual round for residents 55+, so the under-representation of residents 65+ remains. Of sixteen development incidents, thirteen are closed and I-04 is reopened. The two governance issues introduced by the 11 Nov change were accepted by the team lead: passwordless identity selection, under which the account no longer proves who approved, I-13, and a project-wide Reset demo that deletes every issue in the configured Jira project without a named session, I-14.

Contents

Abstract **i**

1 Scope and sources of the records **1**

2 Audit records of agent write operations **3**
2.1 Audit schema . . . 3
2.2 Agent writes in the demonstration run . . . 3
2.3 Coverage check . . . 3
2.4 Hand-offs without a human in between . . . 4

3 Who approved each checkpoint **5**
3.1 Demonstration run: approvals.csv . . . 5
3.2 Team's real approvals . . . 7

4 Bias and fairness check BIAS-1 **8**
4.1 Method . . . 8
4.2 Findings . . . 8
4.3 Actions proposed by the check . . . 10
4.4 Status and team decision . . . 10

5 Guardrail events **11**

6 Incidents and the changes we made **13**
6.1 Summary of PM-tool write paths . . . 15

References **16**

A Approval rows of the demonstration run **17**

B Reset check results **18**

C Audit action counts **19**

D Reproduction commands **20**

E Jira snapshot of the team's real run **21**

1 Scope and sources of the records

This Governance Log belongs to the Harbourline AI PM system of MA6094 Capstone Group 5. It is dated 14 November 2026. The governance owner is Huang Hao, G2510626K; the team lead is Tian Yujia, G2510505D.

The system writes its governance records itself, into the SQLite tables `audit_log`, `approvals` and `handoffs`, see `app/db.py:58-64,81-83`. Keeping these records inside the system, rather than reconstructing them afterwards, follows the "govern" and "map" functions of AI risk management, which ask for documented accountability and traceable decisions (Tabassi, 2023). Two kinds of record are reported here. Table 1-1 lists them, together with the team's real run on its Jira site.

Table 1-1 Sources of the governance records

Source	What it is	Files
Demonstration run, regenerated 14 Nov 2026 on the latest code	One end-to-end session produced by `scripts/governance_demo.py` on a throw-away database. The rows, IDs, hashes and decisions are identical to the first run of 8 Nov 2026; only timestamps, the run ID and the run time changed. It runs in baseline mode, with no LLM calls, against the mock board. Every approval goes through the same server handler as the UI, `server.api_approve`, signed by the seeded **role accounts**. These are demonstration decisions, not the team's real decisions.	In `docs/governance/evidence/`: `audit_log.csv`, 188 rows; `agent_writes.csv`, 137 rows; `approvals.csv`, 10 rows; `bias_check.json`; `change_impact_CE-01.md`; `summary.json`
Reset check, 14 Nov 2026	`scripts/reset_check.py` exercises the start-screen Reset demo on a throw-away database with the mock board and Jira credentials blanked	`docs/governance/evidence/reset_check.json`
Team's real run	The run on the team's Jira site, made from the team's demo machine: Free plan, project `SCRUM`, token of Tian Yujia	Audit and approvals exports are kept on that machine, as the team states. In the repository: a read-only Jira snapshot taken on 14 Nov 2026, `docs/governance/evidence/jira_snapshot_2026-11-14.json`, with 92 issues `SCRUM-283`–`374`, all `ai-generated`, 92 distinct external keys and 18 Blocks links; see Appendix E

Reproduction and token hygiene. The two evidence sets are reproduced with `python3 scripts/governance_demo.py` and `python3 scripts/reset_check.py`; see Appendix D. The export contains no tokens: a scan for token patterns found 0 matches, and all log text passes through `audit.redact()` in `app/audit.py:33–40`.

Structure of this log. Chapter 2 reports the audit records of agent write operations. Chapter 3 shows who approved each of the human checkpoints HC1 to HC7. Chapter 4 covers the bias and fairness check BIAS-1, Chapter 5 the guardrail events including the Reset demo, and Chapter 6 the incidents I-01 to I-16 and the changes made. The appendices list the approval rows, the reset check results, the audit action counts, the reproduction commands and a summary of the read-only Jira snapshot.

2 Audit records of agent write operations

2.1 Audit schema

Each audit row has the fields `id`, `ts`, `run_id`, `actor`, `actor_type`, `action`, `target_type`, `target_id`, `outcome` and `detail`, defined in `app/db.py:61–64`. The actor type is `agent`, `human` or `system`. The outcome is `ok`, `blocked`, `rejected`, `fallback` or `error`, and the detail is JSON. Each agent write calls `audit.log()`, see `app/audit.py:43–49`; PM writes are logged inside the guarded adapter in `app/pmtool.py:356–376`.

2.2 Agent writes in the demonstration run

Table 2-1 groups the 137 agent-write rows of `agent_writes.csv` by agent and write action, with one example per group. Figure 2-1 shows the same rows by agent, together with the outcome of all 188 audit rows.

Table 2-1 Agent writes in the demonstration run, agent_writes.csv, 137 rows

Agent	Write action	Rows	Example: audit id → target
planning	`write plan`	1	#8: 17 scope lines, 20 Sponsor Questions, 12 conflicts, 74 WBS elements, 18 milestones
risk	`write risk register`, `write what-if`	1 + 4	#11: 22 risks; #12: W1 v1
governance	`bias & fairness check written`	1	#16 → BIAS-1, 6 findings
reporting	`write report draft`	6	#18 → EDC-2027-03-31-v1, hash recorded
tooling	`pm create`	92	#31 → mock issue for `ext-HCH-WBS-1`
tooling	`pm link`, Blocks	18	#123 → milestone dependency
tooling	`pm sync applied`	2	#141: 92 created; second sync 0 created / 0 updated / 92 unchanged
change	`mark stale`, `change impact note drafted`	8 + 1	#154 → R01 stale, tags budget, eligible-cost, grant; #173 → CE-01
risk	`risk re-scored (approved)`	1	#175 → R01 20 → 25, **only after HC6**
change	`pm comment`, `pm update` with label	1 + 1	#185–186: comment and `change-ce-01` label on the WBS 1.2.2 item, **only after HC6**

2.3 Coverage check

- Every PM issue has a `pm create` row: 92 issues, 92 rows. Evaluation test T30 checks the same rule: 92 rows for 92 issues in baseline mode, 118 for 118 in LLM mode.
- No agent write to the PM tool happened before HC2 and HC3 were approved: rows #31 onwards follow approvals #2 and #3.

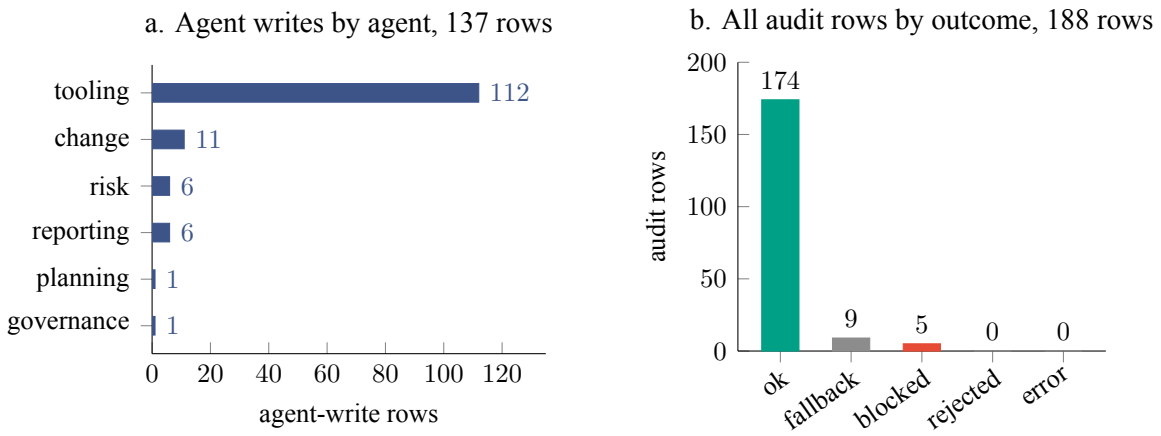


Figure 2-1 Audit rows by agent and by outcome. Panel a: the 137 agent-write rows by agent; tooling covers 92 `pm create`, 18 `pm link` and 2 `pm sync applied`. Panel b: all 188 audit rows by outcome; the 5 `blocked` rows are audit #26 and #148–151, and the 9 `fallback` rows are baseline outputs made without LLM calls. Source: `agent_writes.csv`, `audit_log.csv`.

2.4 Hand-offs without a human in between

The `handoffs` table records four agent-to-agent hand-offs that pass without a human checkpoint: planning → risk, risk → reporting, reporting → critic and change → reporting. Evaluation test T31 checks them.

3 Who approved each checkpoint

The system has seven human checkpoints, HC1 to HC7. Recording who decided what, and refusing decisions from roles that do not hold the authority, is the approval and change-control discipline that project governance asks for (Project Management Institute, 2021). Figure 3-1 places the checkpoints of the demonstration run in order, with the attempts that were refused.

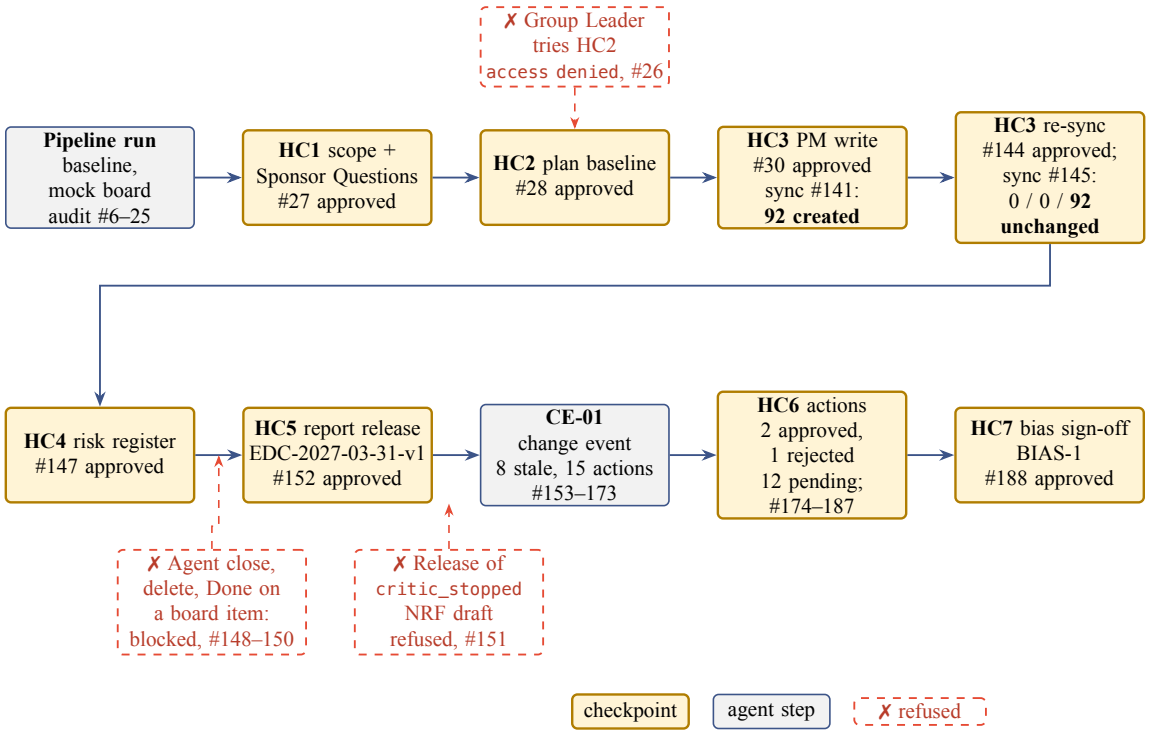


Figure 3-1 Demonstration run in the order of the audit log. Yellow boxes are human checkpoints, with approvals signed by seeded role accounts; dashed orange boxes are attempts refused and logged with outcome `blocked`. The HC3 re-sync shows created / updated / unchanged. Source: `audit_log.csv`, `approvals.csv`, `summary.json`.

3.1 Demonstration run: approvals.csv

The approver is the selected session account. The server takes it from the session and never from a typed name, see `server.py:118–143`. Current passwordless identity selection does not verify the person using that account; **these demonstration approvals are not evidence of a real person's sign-off**. Incident I-13 in Chapter 6 records this. Table 3-1 lists the ten approval rows, and Appendix A gives the full rows of `approvals.csv`.

Refused, as designed.

- The Group Leader account tried to approve HC2. It was refused before anything was recorded, and logged as `access denied: HC2` with outcome `blocked`. This is audit row #26; the check is in `app/auth.py:196–202`.

Table 3-1 Approvals in the demonstration run, approvals.csv

#	Checkpoint	Item approved	Approver account	Decision	Comment
1	HC1 scope + Sponsor Questions	gaps `072aec9495dc`	Project Manager, `project_manager`	approved	scope and Sponsor Questions reviewed
2	HC2 plan baseline	plan `2656fb7c1a17`	Project Manager	approved	WBS, dictionary and milestones baselined
3	HC3 PM write	preview `d252c479f4b9`, 92 create	Project Manager	approved	first sync
4	HC3 PM write	preview `c9ad4f3eb54a`, 92 unchanged	Project Manager	approved	re-sync, idempotency
5	HC4 risk register	register `abb8ad434b95`	Group Leader, `group_leader`	approved	register and what-ifs reviewed
6	HC5 report release	EDC-2027-03-31-v1	Group Leader	approved	released after Critic PASS
7	HC6 change action	CE-01-A01: re-score R01	Project Manager	approved	re-score accepted
8	HC6 change action	CE-01-A04: comment + label WBS 1.2.2	Project Manager	approved	—
9	HC6 change action	CE-01-A14: milestone note	Project Manager	**rejected**	wording to be agreed with the sponsor first
10	HC7 bias sign-off	BIAS-1	Supervisor, `supervisor`	approved	findings accepted

- The Group Leader account tried to release a planted-fault NRF draft whose status was `critic_stopped`. Release was refused, as audit row #151 shows; the check is in `app/reporting.py:295–298`.

12 of the 15 CE-01 actions were left pending. Nothing was applied without HC6.

3.2 Team's real approvals

Table 3-2 maps each checkpoint to the role that may approve it under role-based access control, defined in `app/auth.py:25–57`. In the team's real run, approvals were made with the role accounts on the team's demo machine.

Table 3-2 Team's real approvals

Checkpoint	RBAC role	Named approver	Date
HC1, HC2, HC3, HC6	`project_manager`	Project Manager account	HC2 and HC3 before the Jira write of 14 Nov 2026 16:51 SGT; others on the team machine
HC4, HC5	`group_leader`	Group Leader account	On the team machine
HC7	`supervisor`	Supervisor account	On the team machine

HC1, HC4 and HC7 are recorded and displayed but do not block any code path; this is limitation L4 in §4 of the design document. HC2, HC3, HC5 and HC6 are enforced in code.

4 Bias and fairness check BIAS-1

The governance agent runs a bias and fairness check, BIAS-1, on the resident consultation responses. It looks for skew by subgroup and by data-collection channel, the kind of bias assessment that AI risk management guidance asks for (Tabassi, 2023).

4.1 Method

The method is implemented in `app/governance.py:43–107`:

- Tag each of the 214 consultation responses in `05_Resident_consultation_responses.csv` with themes, using a lexicon in English, Chinese, Malay and Tamil.
- Rank the themes three ways: by raw count, with each age band weighted equally, and with English-only responses.
- Report skew by channel, age and language.

4.2 Findings

Table 4-1 lists the six findings recorded in `docs/governance/evidence/bias_check.json`. Figure 4-1 draws the response counts by age band and channel, and the raw and age-re-weighted ranks of the top four themes, from the same file.

Table 4-1 Bias and fairness findings of BIAS-1

ID	Severity	Finding
B1	high	Online channel gave 160 of 214 responses, 74.8%, and 115 of them came from under-45s. Seabreeze saw the same skew, LL-L06.
B2	high	Residents 65+ gave 25 responses, 11.7%, although they are the main users of the active ageing centre; under-45s gave 121, 56.5%. The estate's age profile is NOT FOUND in the pack, so true representativeness cannot be computed.
B3	high	All 19 non-English responses, zh 10, ms 6 and ta 3, came on paper or at the town hall, 18 of them from residents 45+. An English-only summary would drop 11 of 36 active-ageing comments and 6 of 37 lift/access comments.
B4	medium	Raw top four: Childcare, Communication & trust, Lifts & access, Active ageing. Age-re-weighted top four: Childcare, **Active ageing**, Lifts & access, Communication & trust. The order changes, so both must be reported.
B5	medium	Only 25 distinct comment texts across 214 responses; check for template or campaign submissions before weighting.
B6	low	Mean rating by age band ranges from 2.65 for 35–44 to 3.48 for 18–24. The town hall drew about 40 residents, EDC-4.5, and gave 31 responses.

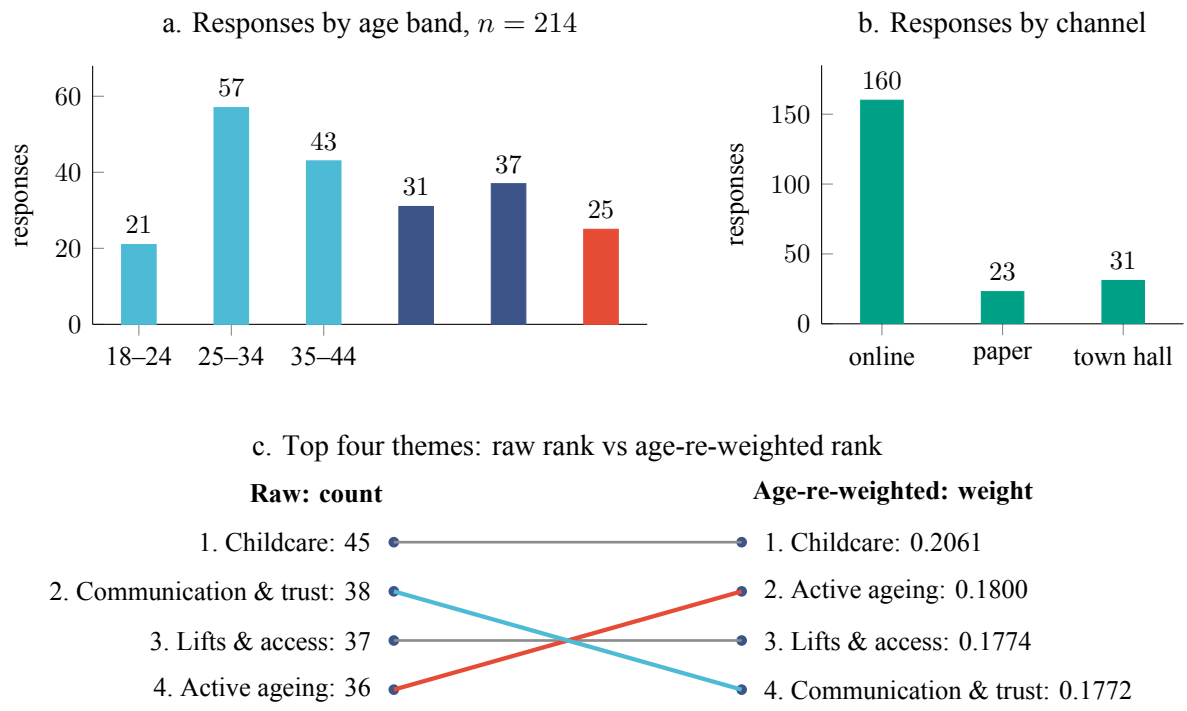


Figure 4-1 Bias check BIAS-1. Panel a: responses by age band, light blue under 45, navy 45–64, orange 65+; residents 65+ gave 25 or 11.7%, under-45s 121 or 56.5%. Panel b: responses by channel; online gave 160 of 214, 74.8%. Panel c: rank of the top four themes by raw count and with each age band weighted equally. Active ageing rises from fourth to second and Communication & trust falls from second to fourth, finding B4. Re-weighted values are the theme weights in the file; theme names are shortened as in finding B4. Source: `bias_check.json`, fields `age`, `channel`, `themes_raw`, `themes_reweighted`, `rank_raw` and `rank_reweighted`.

4.3 Actions proposed by the check

1. Translate and include every non-English comment.
2. Show raw and age-re-weighted priorities side by side.
3. Run a second multilingual paper or door-to-door round for residents 55+; see LL-L06.
4. Have Community Engagement confirm whether the repeated submissions are independent.
5. Hold the consultation summary D15 until HC7.

4.4 Status and team decision

Status. The Supervisor account signed BIAS-1 off in the demonstration run; this is approval #10 in Table 3-1. As noted in Section 3.1, it is a demonstration decision, not the team's.

What the team changed as a result: the team's HC7 decision. The team approved and carried out the proposed actions except the most critical one, action 3, the second multilingual paper / door-to-door round for residents 55+, which was not approved. So non-English comments are translated and included, action 1; raw and age-re-weighted priorities are shown side by side, action 2; repeated submissions are checked, action 4; and the summary D15 was held until HC7, action 5. **The under-representation of residents 65+, finding B2, therefore remains** and must be stated in the consultation summary.

5 Guardrail events

Guardrails are the code-level interlocks that stop an agent, or a person, from taking an action that the governance design does not allow. They work like the circuit breakers and safety interlocks of production systems: they refuse the action and leave a trace, rather than relying on the caller to behave (Nygard, 2018). Table 5-1 lists each guardrail with its evidence, including the two rows for the start-screen Reset demo added with the 11 Nov change. Figure 5-1 shows the safeguards of the Reset demo.

Table 5-1 Guardrails and their evidence

Guardrail	Evidence
Agents never delete, close or move to Done	Demonstration: close, delete and Done attempts on a board item were all blocked, audit #148–150, each logged as `BLOCKED: ⋯` with outcome `blocked`. Eval T23: 4/4 blocked.
`ai-generated` label on every agent item	Enforced at create in `app/pmtool.py:357-358`; read-back reports unlabelled items, see `:531`.
No PM write without HC2 + HC3	Eval T21; `app/pmtool.py:485-486,528-531`.
Change actions only after HC6	15 actions drafted, 3 decided, 12 still pending in the demonstration.
Unknown or unpermitted API call refused	Unknown paths default to the `users.manage` permission, see `server.py:370-378`; denied requests are logged `blocked`. Exception: the identity list and the Reset preview and restore requests are served before any session check, see `server.py:408-415`.
Reset demo cannot run by accident	Needs a fresh preview, the typed project key, or `RESET` on the mock board, and a matching `preview_id`; one reset at a time, see `server.py:49-81`. Reset check: wrong confirmation → 400, stale preview → 409, valid reset → 200, recorded in `reset_check.json` and Appendix B.
Reset demo cannot half-reset	Jira issues are deleted deepest-first with `deleteSubtasks=false`, one audit row per deletion; the first Jira error stops the reset before any local data is cleared, see `app/pmtool.py:413-448`. **Not yet exercised against Jira.**
Audit history survives a reset	Reset check: `audit_log` 140 → 166 rows; `approvals` 2 → 0 and `decisions` cleared, as designed in `app/pipeline.py:83-100`.
Personal tokens never in shared files	`.env` is git-ignored; `.env.example` has no values; text is redacted before logging; no token is in the git history, see §0.1 of `docs/design/00_inventory.md`.

The Reset demo is a human path, not an agent path, and its residual risks are tracked as incident I-14 in Chapter 6: it deletes every issue in the configured Jira project, runs without a session, and clears the `approvals` and `decisions` tables.

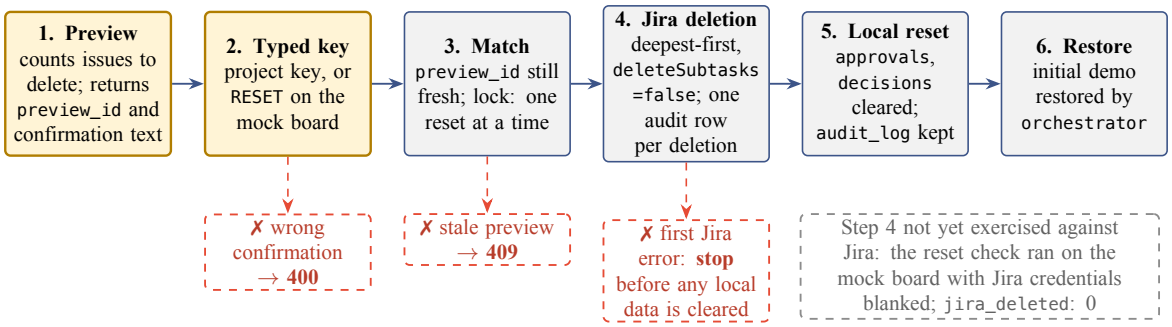


Figure 5-1 Safeguards of the start-screen Reset demo, implemented in `server.py:49–81`, `app/pmtool.py:413–448` and `app/pipeline.py:83–100`. Yellow steps need the person's input; dashed boxes are refusals. The reset check returned 400 for a wrong confirmation, 409 for a stale preview and 200 for a valid reset, and the audit log grew from 140 to 166 rows. Source: `reset_check.json`.

6 Incidents and the changes we made

These are problems found during development and review, from the commit history, code comments and the development sessions of 6–8 Nov 2026 and the documentation review of 14 Nov 2026. Each entry records what happened, how it was found, the change made and the evidence, in the manner of a postmortem record (Beyer et al., 2016).

Table 6-1 lists incidents I-01 to I-16, in the order of the governance log. Thirteen are closed. I-13 and I-14 are **Accepted** by the team lead on 14 Nov 2026, and I-04 was **Reopened** on 11 Nov by I-13.

Table 6-1 Incidents I-01 to I-16 and the changes made

ID	Date	What happened	How it was found	Change made	Evidence	Status
I-01	by 8 Nov	Jira's search index lags behind writes, so a quick re-sync could miss items just created and duplicate them	Jira testing	Lookup = one search plus a consistent bulk read of keys already written; sync map per backend	`app/pmtool.py: 136–140,452–479`; commit `8044577`	Closed; mock: T22
I-02	by 8 Nov	The direction of Jira "Blocks" links is easy to get backwards	Jira testing	First link per session is read back; a reversed link the agent just created is deleted, logged `pm link direction corrected`, and recreated	`app/pmtool.py: 294–317`; commit `8044577`	Closed
I-03	by 8 Nov	Localised Jira sites name issue types differently, e.g. 长篇故事 / 任务 / 子任务 for "Epic / Task / Sub-task" in Chinese, so lookup by English name fails	Jira testing	Types resolved by hierarchy level, with env overrides	`app/pmtool.py: 245–266`; commit `8044577`; Jira snapshot: 6 长篇故事 / 37 任务 / 49 子任务 created	Closed
I-04	by 8 Nov	Checkpoints could be approved under a typed-in name	Governance review	Login with four roles; approvals signed by the session account; refused attempts logged	`app/auth.py:1–11`, `server.py: 118–130`; commit `8044577`	**Reopened** 11 Nov by I-13
I-05	8 Nov	Design-document inventory found the plan, register and reports were hand-authored tables; the LLM only added extras	Design review, Phase 1	Core outputs now LLM-generated, with validators and baseline fallback; origin recorded per agent	commit `9356955`; `app/gen.py`	Closed
I-06	8 Nov	Generated report drafts were STOPPED by the Critic, so the system fell back to templates. Causes: a range fact used in a calculation, "March 2027" with no source date, a computed date, unattributed "on programme"	Audit rows `critic STOP on generated … draft`	Drafting rules added; the full text of failing lines is fed back; revision limit raised from 2 to 3	`app/reporting.py: 151,176–192,237`	Closed; T18 passes in LLM mode
I-07	8 Nov	NRF JSON truncated at 4,000 output tokens, an `llm json parse` error	Audit log	max_tokens raised to 7,000 and reports capped at 22 lines	`app/reporting.py: 173,193`	Closed
I-08	8 Nov	The LLM proposed a re-score as "R01 Grant conditions…", which failed validation. The fallback then applied the **curated re-score rules to unrelated generated risks**, e.g. R22	Change-event test review	Risk ID parsed by regex; curated rules are never applied to a generated register	`app/risk.py: 394–419`	Closed
I-09	8 Nov	Fallback summary was not cached, so an identical re-run spent ∼770 tokens	Run accounting, `runs` table	Summary uses the response cache	`app/reporting.py: 141–147`	Closed; re-run 0 tokens

continued on next page

Table 6-1 continued

ID	Date	What happened	How it was found	Change made	Evidence	Status
I-10	8 Nov	DeepSeek is not on the brief's permitted tool list, brief L210–L212	Design review	Gemini via Google AI Studio is primary and DeepSeek is backup; `HCH_LLM_ORDER=gemini` removes the backup	`app/config.py:27–59`, `app/llm.py:66–115`	Closed: the course instructor accepted the use of DeepSeek, team statement of 14 Nov 2026
I-11	8 Nov	LLM-mode evaluation 27/33: gap recall 77% < 80% target; five checks pin curated IDs	Evaluation	14 Nov: ID-pinned checks rewritten as behaviour checks; LLM mode now 34/34; recall 77–82% kept as limitation L1, above the threshold the instructor confirmed	`docs/evaluation/evidence/`	Closed
I-16	14 Nov	LLM-mode re-run found three real defects: generated risks with figures their sources do not state, an invention rate of 6.9%; CF12 missing from the CE-01 UNRESOLVED list; no re-score proposed for CE-02. It also found one robustness gap: an uncached extra-gaps call broke re-run stability after a planning fallback	Evaluation, `eval_llm_deepseek_2026-11-14_run1_before_fixes.log`	Risk figures traced with one revision round, then dropped; detected conflicts touching the event always listed; one follow-up ask for re-scores; missing Sponsor Question register requested once more; extra-gaps call cached	`app/risk.py:129–170`, `app/change.py:179–188,285–290`, `app/planning.py:296,344–347`	Closed; final LLM run 34/34
I-12	8 Nov	Review findings: HC3 approval not re-verified against the current diff; memory returns a false NOT FOUND on EDC cadence; HC1/HC4/HC7 record only	Design review	Documented as limitations L2–L4	§8 of `docs/design/design_document.md`	Closed 14 Nov 2026: L3 fixed in `app/memory.py`, eval T34; L2 and L4 accepted by the team
I-13	11 Nov	To speed up demonstrations, passwords were removed: the start screen lists the active accounts and anyone can pick one, including admin. Approvals are still recorded under the session account, but the account no longer proves who approved	Documentation review, 14 Nov	None yet. Documented as design limitation L7 and trade-off T7. The password code is in history at commit `8044577`	`app/auth.py:129–148`	**Accepted** by the team lead, 14 Nov 2026: the team alone controls the demo machine; approvals are attributed to role accounts
I-14	11 Nov	The new start-screen Reset demo deletes **every** issue in the configured Jira project, including issues the system did not create. It runs without a session, so the audit log records the deletions under the actor `demo reset` and the restore under `orchestrator`, not a named person. It also clears the `approvals` and `decisions` tables	Documentation review and reset check, 14 Nov	Safeguards in code: preview bound to `preview_id`, typed project key, lock, deepest-first deletion without cascade, stop on first error before local reset; audit log kept. Documented as limitation L8	`server.py:49–81,408–415`; `app/pmtool.py:401–448`; `app/pipeline.py:83–100`; `reset_check.json`	**Accepted** by the team lead, 14 Nov 2026: the configured project `SCRUM` is the team's sandbox and holds only system items; export approvals before a reset
I-15	14 Nov	The 11 Nov change moved code, so many `file:line` references in the three documents pointed at the wrong lines	Documentation review	References re-mapped by diff against `d4a3649` and spot-checked; documents updated	this update	Closed

6.1 Summary of PM-tool write paths

No **agent** wrote to the PM tool outside the HC3-approved path. The only other path that changes the PM tool is the human Reset demo, incident I-14. Every blocked attempt in the demonstration and evaluation runs is in the audit log with outcome `blocked`.

References

Beyer, B., Jones, C., Petoff, J., & Murphy, N. R. (2016). *Site reliability engineering: How google runs production systems*. O'Reilly Media.

Nygard, M. T. (2018). *Release it! design and deploy production-ready software* (2nd ed.). Pragmatic Bookshelf.

Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK guide) and the standard for project management* (7th ed.).

Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (tech. rep. No. NIST AI 100-1). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.AI.100-1

A Approval rows of the demonstration run

Table A-1 reproduces the ten rows of `docs/governance/evidence/approvals.csv`. All rows have the timestamp 2026-11-14T18:00:11+08:00. Every comment begins with the prefix "Demonstration run, scripted", which is omitted below. The role label is the label stored with the approval; the approver is the seeded role account, not a verified person; see Section 3.1.

Table A-1 Rows of `approvals.csv`

id	Checkpoint	Item type, id	Approver	Role label	Decision	Comment, shortened
1	HC1 Scope statement & Sponsor Questions sign-off	gaps, `072aec9495dc`	Project Manager	Team PM lead	approved	scope and Sponsor Questions reviewed
2	HC2 WBS, dictionary & milestone plan baseline approval	plan, `2656fb7c1a17`	Project Manager	Team PM lead	approved	WBS, dictionary and milestones baselined
3	HC3 PM tool write confirmation, diff preview	sync_preview, `d252c479f4b9`	Project Manager	Tool integration owner	approved	sync preview: create 92
4	HC3 PM tool write confirmation, diff preview	sync_preview, `c9ad4f3eb54a`	Project Manager	Tool integration owner	approved	re-sync to show idempotency
5	HC4 Risk register & what-if publication	risk_register, `abb8ad434b95`	Group Leader	Risk owner, team	approved	risk register and what-ifs reviewed
6	HC5 Report release after Critic pass	report, EDC-2027-03-31-v1	Group Leader	Reporting owner, team	approved	EDC update released after Critic PASS
7	HC6 Change-impact actions approval	action, CE-01-A01	Project Manager	Team PM lead	approved	re-score accepted
8	HC6 Change-impact actions approval	action, CE-01-A04	Project Manager	Team PM lead	approved	comment + label the affected PM item
9	HC6 Change-impact actions approval	action, CE-01-A14	Project Manager	Team PM lead	**rejected**	wording to be agreed with the sponsor first
10	HC7 Bias & fairness sign-off on consultation summary	bias_check, BIAS-1	Supervisor	Governance lead	approved	findings B1–B6 accepted; actions to Community Engagement

B Reset check results

Table B-1 gives the results recorded in `docs/governance/evidence/reset_check.json`, generated on 2026-11-14T18:00:12+08:00. Mode: baseline with no LLM, mock board, Jira credentials blanked; `jira_reset_enabled` is `false`, so no Jira deletion was exercised.

Table B-1 Reset check: table counts before and after the reset

Table / count	Before	After
PM issues	92	0
`approvals`	2	0
`decisions`	0	0
`audit_log`	140	166
users	4	4

Check	Status code	Response
Preview	—	Jira enabled: false; Jira count 0; mock count 92; confirmation text `RESET`
Wrong confirmation	400	“Preview the reset and type RESET to confirm”
Stale `preview_id`	409	“Project data changed since the preview; preview the reset again”
Valid reset	200	reset: true; Jira deleted: 0
Audit log kept	—	true
Reset audit row	—	system actor `orchestrator`, action `initial demo restored`, outcome `ok`

C Audit action counts

Table C-1 counts the 188 rows of `docs/governance/evidence/audit_log.csv` by action and actor type. The 169 agent rows match `counts.by_action` in `summary.json`; the 19 system and human rows are computed directly from `audit_log.csv`.

Table C-1 Audit rows by action, audit_log.csv, 188 rows

Action	Actor type	Rows
pm create	agent	92
pm link	agent	18
output generated (baseline)	agent	9
handoff	agent	9
mark stale	agent	8
write report draft	agent	6
critic critic_passed	agent	6
write what-if	agent	4
pm sync applied	agent	2
pm readback	agent	2
pm preview (read-only)	agent	2
write risk register	agent	1
write plan	agent	1
risk re-scored (approved)	agent	1
pm update	agent	1
pm comment	agent	1
change impact note drafted	agent	1
change event received	agent	1
bias & fairness check written	agent	1
BLOCKED: transition to Done	agent	1
BLOCKED: delete	agent	1
BLOCKED: close	agent	1
user created	system	4
ingest	system	1
run start: full	system	1
run end: full	system	1
checkpoint HC1 approved	human	1
checkpoint HC2 approved	human	1
checkpoint HC3 approved	human	2
checkpoint HC4 approved	human	1
checkpoint HC5 approved	human	1
checkpoint HC6 approved	human	2
checkpoint HC6 rejected	human	1
checkpoint HC7 approved	human	1
access denied: HC2	human	1
release refused	human	1
Total		**188**

By outcome, the 188 rows are 174 `ok`, 9 `fallback` and 5 `blocked`, namely audit #26 and #148–151; no row has outcome `rejected` or `error`. The HC6 rejection of CE-01-A14 is recorded as the action `checkpoint HC6 rejected` with outcome `ok`.

D Reproduction commands

Both evidence sets are regenerated on a throw-away database against the mock board; the reset check also blanks the Jira credentials.

```
# Demonstration run: audit_log.csv, agent_writes.csv, approvals.csv,
# bias_check.json, change_impact_CE-01.md, summary.json
python3 scripts/governance_demo.py

# Reset check: reset_check.json, mock board, Jira credentials blanked
python3 scripts/reset_check.py
```

The team’s real records are exported with the Governance export in the UI.

E Jira snapshot of the team's real run

Table E-1 summarises `docs/governance/evidence/jira_snapshot_2026-11-14.json`, taken on 14 Nov 2026 with read-only requests and no write. The Jira Cloud Free plan is a team statement.

Table E-1 Read-only Jira snapshot, 14 Nov 2026

Field	Value
Project; plan	`SCRUM`; Jira Cloud Free, team statement
Issues; key range	92; `SCRUM-283` to `SCRUM-374`
Created between	2026-11-14 16:51:09 and 16:52:10 SGT
Issue types and hierarchy level	长篇故事 6 at level 1, 任务 37 at level 0, 子任务 49 at level −1
Status	all 92 `Idea`
With `ai-generated` label	92
With external-key label; distinct keys	92; 92
With parent	68
Milestone items	18
Issue links	18 Blocks links; 36 link ends seen
`change-*` labels	none